$$OM = 12\sqrt{3} \qquad \ldots(2)$$

From (1) and (2) we have, $\dfrac{h-12}{\sqrt{3}} = 12\sqrt{3}$

$$\Rightarrow \quad h - 12 = 12\sqrt{3} \times \sqrt{3} \quad \text{we get, } h = 48$$

Hence, the required height of the cable tower is 48 m.

Example 6.32 A pole 5 m high is fixed on the top of a tower. The angle of elevation of the top of the pole observed from a point 'A' on the ground is 60° and the angle of depression to the point 'A' from the top of the tower is 45°. Find the height of the tower. $(\sqrt{3} = 1.732)$

Solution Let BC be the height of the tower and CD be the height of the pole.

Let 'A' be the point of observation.

Let $BC = x$ and $AB = y$.

From the diagram,

$\angle BAD = 60°$ and $\angle XCA = 45° = \angle BAC$

In the right angled ΔABC, $\tan 45° = \dfrac{BC}{AB}$

$$\Rightarrow 1 = \frac{x}{y} \Rightarrow x = y \qquad \ldots(1)$$

In the right angled ΔABD, $\tan 60° = \dfrac{BD}{AB} = \dfrac{BC + CD}{AB}$

$$\Rightarrow \sqrt{3} = \frac{x+5}{y} \Rightarrow \sqrt{3}\, y = x + 5$$

we get, $\sqrt{3}\ x = x + 5$ [From (1)]

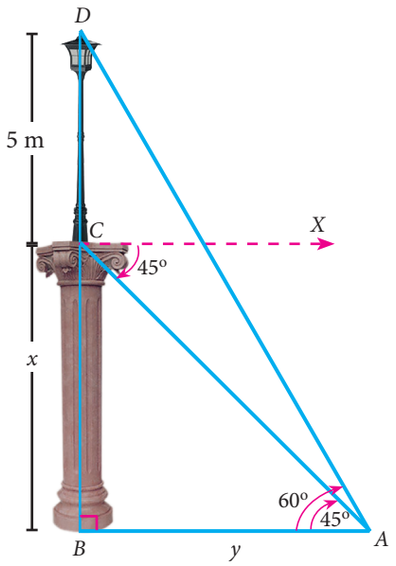


Fig. 6.28

$$x = \frac{5}{\sqrt{3}-1} = \frac{5}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{5(1.732+1)}{2} = 6.83$$

Hence, height of the tower is 6.83 m.

Example 6.33 From a window (h metres high above the ground) of a house in a street, the angles of elevation and depression of the top and the foot of another house on the opposite side of the street are θ_1 and θ_2 respectively. Show that the height of the opposite house is $h\left(1 + \dfrac{\cot\theta_2}{\cot\theta_1}\right)$.

Solution Let W be the point on the window where the angles of elevation and depression are measured. Let PQ be the house on the opposite side.

Then WA is the width of the street.

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